

Human Immunodeficiency Virus Infection and the Kidney

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INTRODUCTION

According to the United Nations acquired immunodeficiency syndrome (AIDS) program, an estimated 38 million individuals had human immunodeficiency virus (HIV)-1 in 2021, including 1.7 million with newly acquired infections. In 2019 an estimated 0.69 million deaths were attributed to HIV-related illness. In 2019 an estimated 25 million individuals were receiving antiretroviral therapy, representing 67% of all HIV-positive (HIV+) individuals. In 2019 around 150,000 children aged 0 to 9 years were newly infected with HIV. This brings the total number of children aged 0 to 9 years with HIV to 1.1 million (<https://data.unicef.org/topic/hivaids>). In the United States, an estimated 1.2 million individuals were HIV+ in 2019 and approximately 2000 patients were receiving renal replacement therapy (RRT; dialysis or kidney transplant) for HIV-associated kidney disease.

HIV-1 infection is associated with glomerular and tubulointerstitial disease, and patients with HIV infection are at risk for nephrotoxicity from antiretroviral therapy¹ and from other medications such as anti-biotics and nonsteroidal antiinflammatory agents (NSAIDs).

Acute kidney injury (AKI), interstitial nephritis from medication and opportunistic infections, and coinfections with hepatitis B virus (HBV) and hepatitis C virus (HCV) are relatively common. Advances in treatment are to be celebrated but also mean that people with HIV are more likely to experience complications such as atherosclerosis, metabolic syndrome, type 2 diabetes, and end-stage kidney disease (ESKD). HIV-2 can cause immunodeficiency but rarely causes kidney disease. In this chapter, we use HIV to refer to HIV-1.

EPIDEMIOLOGY OF HIV CHRONIC KIDNEY DISEASE

High-Income Countries

There are relatively few data on the worldwide prevalence of chronic kidney disease (CKD) associated with HIV infection. In the European EUROSIDA observational study, only 3% had estimated glomerular filtration rate (eGFR) less than 60 mL/min/1.73 m² and only 0.3% had eGFR less than 30 mL/min/1.73 m².² In the United States, the incidence (absolute counts) of ESKD attributed to HIV continues to decline (from 893 individuals in 2006 to 525 in 2011).³

The United States Renal Data System (USRDS) reports that for the period from 2006 to 2010, there were 3834 cases of AIDS nephropathy (0.7% of total ESKD); 86% of the patients were Black and 15% were White. The unadjusted relative risk for ESKD was 4.7 for Black patients compared with Whites. This relative risk is less than previously estimated for HIV-associated nephropathy (HIVAN) and supports the idea that the distribution of kidney diseases associated with HIV infection is shifting. Patients reaching ESKD are also living longer.

Among those who started dialysis in 2009 to 2011, mortality declined by 64% compared with 1988 to 1992 but remained nearly threefold higher than other dialysis patients.³

Lower- and Middle-Income Countries

Sub-Saharan Africa is home to approximately 70% of the global infections, with 26 million HIV+ people. In 2018, 470,000 people died with HIV in Africa. Globally, 91% of HIV+ children live in Africa. Of HIV+ individuals in sub-Saharan Africa, an estimated 54% have access to combination antiretroviral therapy (cART).

Prevention of mother-to-child transmission programs have resulted in a dramatically decreased number of childhood HIV cases. This condition may cease to exist in time with this preventive strategy.⁴ In cases where there has been perinatal transmission, the use of highly active retroviral therapy (HAART) has also largely prevented HIV-associated kidney disease and thus decreased the mortality rate.

Adolescents may still acquire HIV infection, and thus it is important to focus on HIV prevention in this age group. Females aged 15 to 24 are the group with the highest rate of new HIV infection in sub-Saharan Africa.⁵

Compared with high-income countries, access to RRT is frequently limited in low- and middle-income countries. The prevalence of CKD, as assessed by the presence of proteinuria or eGFR, ranges widely in Africa, from 6% to 48.5%.⁶ This wide variation may be partly ascribed to differences in study design, study population, and the definitions used for CKD. Based on single-center studies that used proteinuria as a marker of CKD in HIV+ subjects, the prevalence of CKD in HIV+ subjects ranges from 1% to 6% in Brazil to 20% in Iran and 27% in India.

In South Africa, HIVAN is the dominant kidney disease among HIV+ individuals and is documented in up to 80% of kidney biopsy specimens, which is consistent with findings in the United States in the pre-cART era. As in high-income countries, other kidney causes are also observed, including HIV-associated immune complex kidney disease (HIVICK) from postinfectious and other causes, membranous nephropathy, and interstitial nephritis. Some kidney biopsies show nonspecific changes (e.g., mesangial hyperplasia and chronic active interstitial nephritis) that do not fulfill the diagnostic criteria for HIVAN but do show regression on cART. The latter finding suggests that the diagnostic criteria for HIV-associated CKD may need to be broadened.

HIV-ASSOCIATED NEPHROPATHY

HIV infection is associated with various glomerular disorders (Table 58.1). The classic glomerulopathy of HIV infection is collapsing

TABLE 58.1 Kidney Diseases Associated With Human Immunodeficiency Virus (HIV) Infection or Its Treatment

	Entity	Frequency	Associations
Glomerular	HIV-associated nephropathy	Common	African descent; often advanced HIV disease
	HIV-associated immune complex kidney disease	Common	All ethnic backgrounds
	Thrombotic microangiopathy	Uncommon	
	Membranoproliferative glomerulonephritis ± cryoglobulin-associated vasculitis	Rare	Hepatitis C; enfuvirtide
	Membranous nephropathy	Uncommon	Hepatitis B
	Fibrillary and immunotactoid glomerulopathies	Rare	
	Amyloid nephropathy (AA type)	Rare	
Tubular	Minimal change nephropathy	Rare	Nonsteroidal antiinflammatory medication
	Acute kidney injury	Moderately common	Aminoglycosides, cidofovir, foscarnet
	Proximal tubule injury (Fanconi syndrome)	Moderately common	Tenofovir disoproxil, adefovir, cidofovir, didanosine
	Diabetes insipidus	Uncommon	Amphotericin, tenofovir disoproxil, didanosine, abacavir
	Chronic tubular injury	Moderately common	Amphotericin, cidofovir, adefovir, tenofovir disoproxil
	Crystal nephropathy	Uncommon	Indinavir, atazanavir, sulfadiazine, ciprofloxacin, intravenous acyclovir
Interstitial	Interstitial nephritis	Uncommon	Allergy to β -lactam, sulfonamides, ciprofloxacin, rifampin, proton pump inhibitor, allopurinol, phenytoin; also causes of crystal nephropathy listed previously
			BK virus Immune reconstitution inflammatory syndrome; generally advanced disease; after initiation of antiretroviral therapy

glomerulopathy (CG), also known as collapsing focal segmental glomerulosclerosis (FSGS) and as HIVAN.

Etiology and Pathogenesis

HIVAN occurs during both the acute and chronic phases of HIV infection; in the latter case, it is associated with higher viral RNA levels and lower CD4 T-lymphocyte counts. Studies from human kidney biopsies have shown that HIV can infect glomerular and tubular cells, setting up a chronic and possibly latent infection.⁷ Specific HIV accessory proteins such as Vpr and Nef can damage kidney cells independent of direct infection. In transgenic mice, the expression of HIV accessory proteins Vpr or Nef in podocytes is associated with progressive CKD, loss of podocyte differentiation markers, and features suggestive of human CG and FSGS.⁸ These transgenic mouse experiments also suggest that HIV gene products are toxic to tubular epithelial cells, resulting in both apoptosis and inhibition of cell proliferation.

Host factors also determine susceptibility to HIVAN, especially in individuals of African descent compared with those of European or Asian descent. Apolipoprotein L1, encoded by *APOL1* on chromosome 22, is a component of high-density lipoprotein particles. *APOL1* variants have been identified as a major cause of the predilection for chronic glomerular disease that characterizes populations of sub-Saharan African descent.^{9,10} The presence of two *APOL1* codon-changing variants confer an odds ratio of 29 for HIVAN in the United States¹¹ and 89 in South Africa,¹² a strikingly large effect for a complex disease. Homozygosity, or dual heterozygosity for two risk alleles (termed G1 and G2; G1 is actually a haplotype, but the term allele is used here for simplicity), is observed in 72% of African Americans and South Africans with HIVAN, compared with only 12% of African American controls and 3% of South African controls. Among South Africans, a single copy of the *APOL1*-G1 variant confers an odds ratio [OR] of 21 for HIVAN. These variants appear to protect individuals from African sleeping sickness caused by *Trypanosoma brucei rhodesiense*, and this may explain the rapid rise in allele frequency in Africa, particularly West African populations, that has occurred in the past 40,000 years.⁹

In a U.S. study, among HIV+ individuals with proteinuria and two *APOL1* risk variants, 76% had FSGS or CG and had a nearly threefold increased risk of progression to ESKD compared with those with other biopsy findings. *APOL1* high-risk genotype individuals manifested more CG and greater extent of glomerulosclerosis, tubular atrophy, and interstitial fibrosis.¹³ HIV+ women with CKD with two *APOL1* risk alleles had more severe proteinuria, more rapid decline in eGFR, and a 1.7- to 3.4-fold greater risk for incident CKD.¹⁴

The particular association between HIV infection and *APOL1* kidney disease is likely a consequence of HIV infection promoting systemic interferon production, which, in turn, promotes expression of the *APOL1* gene.

At least seven mechanisms have been identified to explain the toxic effect of the *APOL1* variants on kidney cells. Evidence for these seven mechanisms have been shown in molecular studies, cell culture, or animal models. Most mechanisms are envisioned to be acting chiefly on podocytes to produce glomerular disease.

1. Increased flux through plasma membrane cation channels, either Na⁺ and Ca²⁺ or K⁺, the latter activating stress kinases and leading to cell swelling.¹⁵
2. Increased generation of interleukin (IL)-1 β , leading to pyroptosis and podocyte loss.¹⁶
3. Disruption of an *APOL1*-mir193 axis, leading to dysfunction of the endoplasmic reticulum and protein production.¹⁷
4. Mitochondrial stress, leading to podocyte energy deficits.¹⁸
5. Activation of protein kinase R, leading to reduced protein synthesis.¹⁹
6. Formation of a tripartite complex of *APOL1*, soluble urokinase-type plasminogen activator receptor (suPAR), and α v β 3 integrin, resulting in integrin activation, altering podocyte-matrix interactions, and impairing podocyte function.²⁰
7. Alterations in *APOL1*/*APOL3* interactions.²¹

Some of these mechanisms have shown only or largely in cell culture studies or animal model, and their clinical relevance remains to be established. Novel therapies for *APOL1* kidney disease are currently in clinical trials.

Clinical Manifestations of HIVAN

Patients with HIVAN typically present with proteinuria and reduced eGFR. Some patients present with edema, although edema is less common than with other conditions associated with nephrotic range proteinuria, suggesting a defect in tubular sodium handling. Imaging studies may reveal increased kidney size despite reduced eGFR and, in some cases, increased echogenicity; this unusual feature is shared with diabetic nephropathy and amyloid nephropathy. However, no ultrasound features predict the kidney histology, and kidney biopsy is required for diagnosis.

Pathology of HIVAN

HIVAN is indistinguishable from idiopathic collapsing FSGS (CG), with pathologic changes in glomeruli (proliferation and dysregulation of podocytes or podocyte stem cells, together with glomerular collapse), tubular injury (both acute and chronic injury, sometimes with microcystic tubular changes), and interstitial changes (chronic inflammation, fibrosis; Fig. 58.1). On electron microscopy (EM), both HIVAN and idiopathic collapsing FSGS may manifest tubuloreticular inclusions within dilated endosomal compartments within glomerular endothelial cells, which are believed to be markers for interferon-mediated injury. Some patients with progressive loss of kidney function (e.g., from HIVICK or interstitial nephritis) may develop postadaptive FSGS from the consequences of glomerular hyperperfusion and hyperfiltration.

Diagnosis and Differential Diagnosis

Indications for kidney biopsy include AKI without clear associated cause, especially with a nephritic sediment; nephrotic proteinuria; laboratory evidence of thrombotic microangiopathy (TMA); and unexplained CKD. In the past, some clinicians argued that nephrotic proteinuria in an individual of African descent is likely to be HIVAN, which is a form of collapsing FSGS or CG, and that a biopsy is not necessary. The widespread use of antiretroviral therapy (ART) in the current era makes this decision less tenable, and kidney biopsy is often indicated to guide prognosis and therapy, particularly when substantial proteinuria is present.

With the advent of ART, a wider range of HIV-associated kidney diseases is now seen. Rather than the initial pattern of HIVAN in individuals of African descent and HIVICK among all ethnic backgrounds, HIV-associated kidney diseases now includes FSGS, diabetic nephropathy, and arterionephrosclerosis (the term is used for the histopathology that is seen in some APOL1-associated kidney disease but also with aging). In patients with HIV and hepatitis, membranoproliferative glomerulonephritis associated with HCV should be considered. Other diagnoses include membranous nephropathy (associated with HBV infection, or presumably idiopathic, but possibly related to the polyclonal B cell expansion typical of HIV disease) and amyloidosis. Chronic tubular injury is usually associated with tenofovir disoproxil fumarate (TDF) therapy but may occur with other ART; when early diagnosis leads to cessation of the offending medication, CKD is

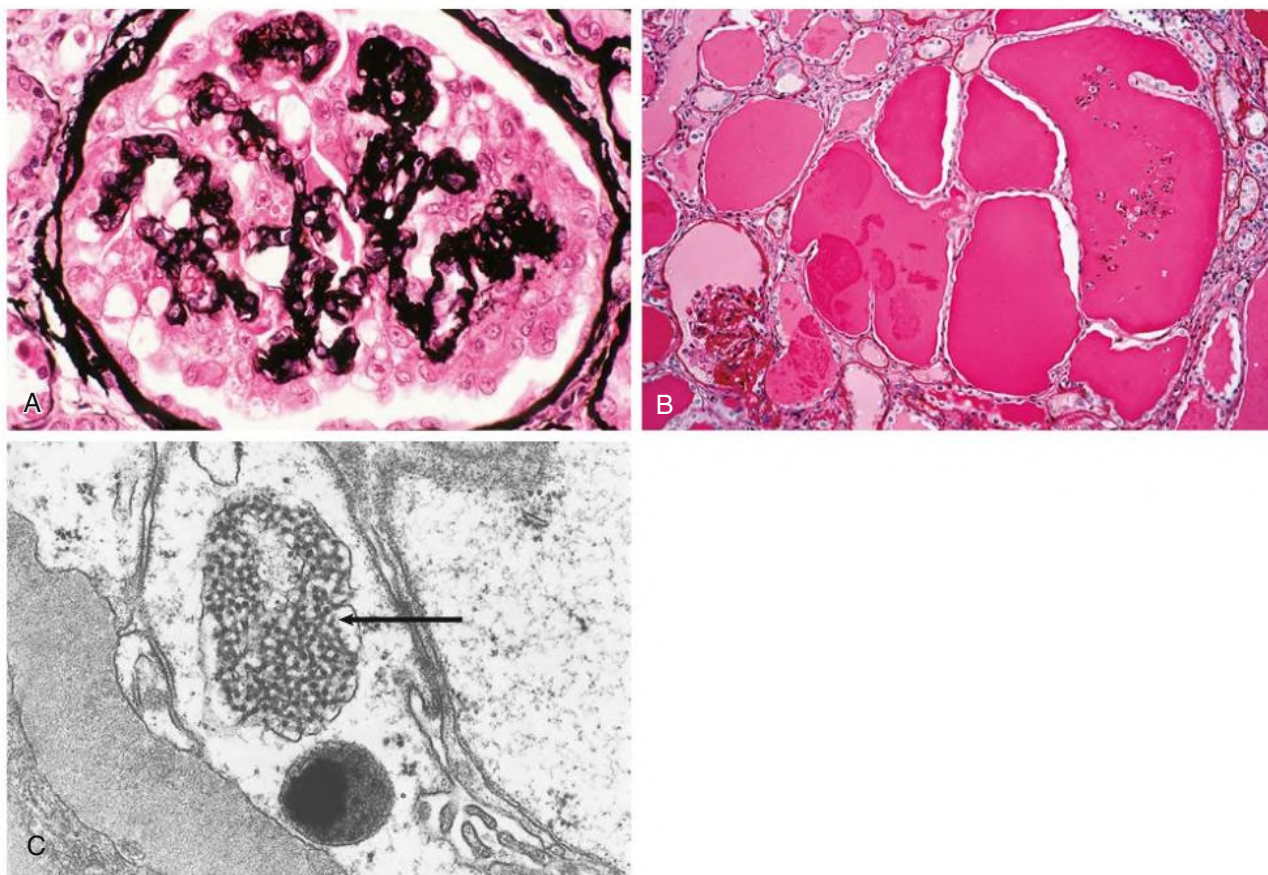


Fig. 58.1 Human Immunodeficiency Virus–Associated Nephropathy. (A) A globally collapsed glomerulus shows marked podocyte hypertrophy and hyperplasia (Jones methenamine silver). (B) At low power, the kidney parenchyma contains abundant tubular microcysts with proteinaceous casts. The glomerulus is collapsed with dilated urinary space (periodic acid–Schiff stain). (C) The glomerular endothelial cell pictured here contains a large intracytoplasmic tubuloreticular inclusion (“interferon footprint”; arrow) composed of interanastomosing tubular structures within a dilated cisterna of endoplasmic reticulum (electron micrograph).

usually averted. Despite the plethora of diagnoses, HIVAN remains the leading cause of ESKD in patients with HIV in the United States.

The diagnoses of HIVAN and HIVICK, and other diagnoses previously listed, are made by kidney biopsy. In some cases, nephrotic proteinuria and low CD4 T-lymphocyte count in individuals of African descent is sufficiently suggestive of HIVAN that some clinicians may opt not to biopsy the kidney. Plasma HIV RNA of less than 400 copies/mL at the time of presentation suggests another kidney disease. Among individuals of African descent, other possible diagnoses include primary or secondary FSGS (although a pathogenic role for HIV cannot be excluded in conferring susceptibility) and arterionephrosclerosis (see later discussion). In all individuals, diabetic nephropathy and HIVICK should be considered (see later discussion and Table 58.1).

Treatment of HIVAN

The HIV Medicine Association of the Infectious Diseases Society of America provides clinical practice guidelines for the management of HIV disease, with the most recent update published in 2020.²² All HIV+ individuals should receive cART¹ beginning at the time of diagnosis. This may include a three-drug combination or two-drug combination that includes an integrase strand transfer inhibitor. The presence of kidney disease does not affect this recommendation.²³ In the era before cART was available, HIVAN progressed rapidly, with patients reaching ESKD in months to a few years. The marked decline in HIV-associated ESKD in the United States after the introduction of cART in the mid-1990s suggests that effective control of viral replication, achievable with cART, prevents HIVAN. Retrospective studies also suggest that cART treatment of HIVAN may prolong kidney survival.²⁴

A growing number of prospective and retrospective studies have demonstrated rapid resolution or reduction in proteinuria within 6 months of commencing ART. The Development of Anti-Retroviral Therapy in Africa (DART) study, conducted in Uganda and Zimbabwe, demonstrated improvement of kidney function of 2 to 6 mL/min/1.73 m² after 4 to 5 years of ART.^{25,26} Other studies suggest that ART is most effective when it is initiated before the onset of severe kidney disease.^{27–31} In a few cases, regression of histologic lesions with cART has been demonstrated.^{32,33} According to a French study, the pattern of kidney disease has changed since the introduction of ART, with a relative decrease in HIVAN and emergence of classic FSGS as the commonest cause of glomerular disease.³⁴ In this study, HIVAN occurred more frequently in Black patients with severe immunodeficiency and severe kidney failure, whereas FSGS patients were older, more likely to have received cART, more frequently had cardiovascular risk, and had more severe interstitial fibrosis on kidney biopsy.

Treatment of HIV-associated kidney disease also includes standard therapies for CKD, including control of BP and the use of an angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB), combined with a thiazide diuretic, low sodium diet, and weight loss if obese. An aldosterone antagonist such as spironolactone or eplerenone, used in low doses, may potentiate the anti-proteinuric effect of this regimen but requires monitoring to avoid hyperkalemia. A prospective and controlled, but nonrandomized, trial of HIVAN demonstrated improved kidney survival with ACE inhibitor therapy.³⁵ Glucocorticoids have shown benefit for HIVAN in a prospective and several retrospective studies, as reviewed recently.³⁶ Nevertheless, although there is no consensus, these medications should only be considered in HIVAN and HIVICK when viral replication is suppressed with cART and proteinuria or progressive kidney function decline continues. BP targets should be treated to achieve a systolic BP of less than 120 mm Hg consistent with the 2020 Kidney Disease Improving Global Outcomes (KDIGO).¹

Natural History of HIVAN

Progression of CKD is more rapid in HIV+ individuals of African descent, probably reflecting the effect of the *APOL1* variants. Patients with HIVAN may progress to ESKD despite cART, but the use of these antiretroviral therapies may slow or even halt progression. Predictors of progression are a high degree of chronic damage on histology as well as severe kidney dysfunction at baseline, high-grade proteinuria, and presence of many sclerotic glomeruli.^{37,38}

OTHER GLOMERULAR DISORDERS

HIV-Associated Immune Complex Disease

In populations of European and Asian ancestry, the most common glomerular disease associated with HIV disease is immune complex disease (reviewed in Cohen and Kimmel³⁹; case series have been reported from high-income countries, including the United States,^{40,41} the United Kingdom,⁴² and France,³⁴ and from low-income countries, including Thailand⁴³ and India⁴⁴). HIVICK is also seen in populations of African descent, although much less frequently than HIVAN, and most lack two *APOL1* risk alleles. Compared with those with HIVAN, HIVICK subjects tend to have lower HIV viral copy number and are more likely to have diabetes and hypertension.⁴⁵

HIVICK can take various forms, including postinfectious glomerulonephritis, membranous nephropathy, and a lupus-like histologic pattern.⁴⁶ Patients with HCV coinfection may manifest a membranoproliferative glomerulonephritis (MPGN) pattern (see Chapter 22). Some patients may have only mesangial immunoglobulin A (IgA) deposits (especially those with microhematuria and nonnephrotic proteinuria), resembling idiopathic IgA nephropathy. Hence, the diagnosis of HIVICK may be fraught with challenges for the pathologist, who must exclude other immune complex disease.

HIVICK may be clinically indistinguishable from HIVAN, although hematuria may be more marked, and some patients present with lower levels of proteinuria. Kidney biopsy findings may vary from MPGN to focal or diffuse proliferative glomerulonephritis with endocapillary proliferation (Fig. 58.2). In some cases, changes consistent with HIVAN may also be present, including tubuloreticular inclusions within glomerular capillary endothelial cells. Immune deposits may be in mesangial, subendothelial, or subepithelial locations and may include IgG and IgM or IgG, IgM, and IgA (so-called “full house deposits”), often with C3. These forms resemble lupus nephritis, although lupus serologic test results are typically negative. If lupus serologies are positive, then in most cases the individual has lupus and HIV-1 infection, although early case reports suggest that some patients with HIV-1 infection may have false-positive lupus serology tests.⁴⁷

The pathogenesis of HIVICK is not well understood and may involve various mechanisms. It is not known whether complexes form locally within the glomeruli or are passively trapped as from the glomerular microcirculation. In some cases, immune complexes include HIV antigens. Other cases may represent generalized polyclonal B-cell expansion that accompanies HIV disease.

An important issue is whether some or perhaps many HIVICK cases are not because of HIV-1 infection per se but instead are examples of idiopathic glomerular immune disease, in some cases possibly arising as a consequence of the immune reconstitution syndrome after institution of ART. Membranous nephropathy may appear after immune reconstitution, and such patients often have antibodies against the phospholipase-A2 receptor (PLA2R), the most common antigen associated with idiopathic membranous nephropathy.

HIVICK is often associated with progressive loss of eGFR. Foy and colleagues in the United States found that at 2 years after diagnosis, 70% of HIVAN cases and 32% of HIVICK cases had reached

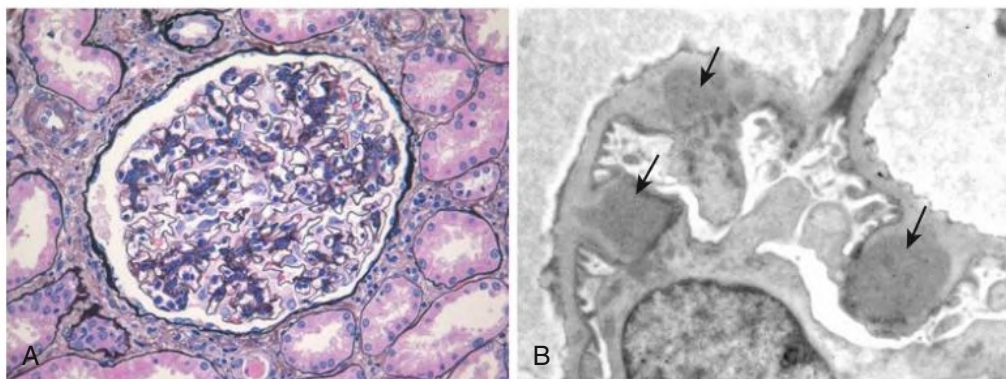


Fig. 58.2 Human Immunodeficiency Virus (HIV)-Associated Glomerulonephritis. This HIV-positive individual presented with microhematuria and nonnephrotic proteinuria with normal kidney function. Kidney biopsy shows mild mesangial expansion with mesangial and paramesangial deposits, together with varying numbers of subepithelial-based deposits (arrows), resulting in the “ball-and-cup” appearance. (Courtesy Professor Stewart Goetsch, University of the Witwatersrand, Johannesburg, South Africa.)

ESKD. Further, the use of cART or the attainment of viral suppression was not associated with lower risk of ESKD in HIVICK, whereas use of cART was associated with reduced likelihood of progression with HIVAN.⁴⁵ In contrast, a UK study reported a slower rate of progression of HIVICK to ESKD, but these authors considered HIV-associated IgA nephropathy to be a distinct entity and with a risk for progression that is intermediate between HIVAN and HIVICK.⁴²

OTHER GLOMERULAR DISORDERS

Thrombotic microangiopathy may occur with HIV infection and is associated with low CD4 cell counts, high viral burden, and AIDS.⁴⁸ Fibrillary glomerulonephritis, immunotactoid glomerulopathy, and amyloidosis have also been reported in HIV infection, but a causal role for HIV infection has not been established.

SYSTEMIC INFLAMMATION AND ARTERIONEPHROSCLEROSIS

A paradigm shift has emerged over the past two decades, with the recognition that long-term suppression of HIV replication dramatically improves survival but at the cost of chronic inflammation and premature aging.⁴⁹ Peripheral inflammatory markers correlate with Framingham risk scores and Veteran Aging Cohort Scores,⁵⁰ and hypertension is common.⁵¹ The causes of chronic inflammation in HIV disease are complex. Chronic immune activation is associated with lymphocyte depletion, manifesting as reduced numbers of CD4⁺ and CD8⁺ naïve T cells. Alterations in the intestinal microbiome and microbial translocation may contribute.⁵²

Inflammation promotes disease processes that resemble premature aging (including atherosclerosis) and manifests in the kidney as arterionephrosclerosis. This disease involves systemic arterial intimal thickening, medial hypertrophy, duplication of the internal elastic lamina, and global glomerulosclerosis. These afferent arteriolar lesions predispose to glomerulosclerosis and hypertension by altering kidney autoregulation. Arterionephrosclerosis was traditionally ascribed to hypertension, but it is now apparent that other mechanisms often contribute. Arterionephrosclerosis is associated with genetic mutations (e.g., *APOL1*), metabolic disorders (e.g., diabetes, obesity, hyperlipidemia, metabolic syndrome, and HIV infection), and chronic inflammation (smoking, oxidative stress, hemodynamic shear stress, and renin-angiotensin system activation).⁵³ A U.S. study of HIV-1 patients reported that isolated arterionephrosclerosis was the most common single pathology,⁵⁴ and this trend has become stronger as more HIV+

individuals have achieved viral suppression and are living longer while taking ART for longer durations.

TUBULAR DISORDERS

HIV infection is associated with tubular disorders, often because of medications (see Table 58.1), the most common being tenofovir disoproxil fumarate (TDF). AKI most commonly occurs in outpatient settings, with prekidney causes (extracellular volume contraction, or reduced kidney blood flow because of nonsteroidal anti-inflammatory agents [NSAIDs]) and postrenal causes (including obstruction as a result of prostate disease, kidney stone, or crystalluria). Kidney causes of AKI include tubular injury from medication, particularly TDF; interstitial nephritis because of NSAIDs or antibiotics; and less commonly, acute presentations of glomerular disease. Risk factors for HIV-associated AKI include male sex, sub-Saharan African ancestry, low CD4 count, high viral load, diabetes, CKD, and hepatic disease.⁵⁵ Chronic proximal tubular injury manifests as various components of Fanconi syndrome: glycosuria, phosphaturia (sometimes associated with clinically significant hypophosphatemia), uricosuria, tubular proteinuria (predominantly low-molecular-weight proteins), and aminoaciduria. Diabetes insipidus may also be present, indicating that the distal nephron is also involved.

NEPHROTOXICITY AND DOSING OF ART AND ASSOCIATED DRUGS

Current Antiretroviral Drug Regimens and Kidney Toxicity

As of 2021, there are 25 antiretroviral medications and 24 antiretroviral medication combination formulations approved for use in the treatment of HIV disease in the United States. Recommendations for first-line therapy of HIV in adults include a combination of three medications. These are TDF, a nucleotide reverse transcriptase inhibitor (NRTI); either lamivudine or emtricitabine, nucleoside reverse transcriptase inhibitors, also abbreviated NRTI; and efavirenz, a nonnucleoside reverse transcriptase inhibitor (NNRTI). The difference between a nucleotide and a nucleoside is that the latter must be phosphorylated to become active. To underscore its ubiquity, TDF is a component of four of the six regimens currently recommended for treatment-naïve patients and hence widely used globally. TDF nephrotoxicity is discussed later. By contrast, NNRTIs have an excellent kidney safety profile.

In addition to these two drug classes (NRTI, NNRTI), there are six other classes of antiretroviral medications currently in clinical use

TABLE 58.2 Adjustments in Drug Dosing for Selected Antiretroviral Medications

Medication Class	Medication	Adjustment for GFR	Dosing in Dialysis
NRTI	Abacavir	No change	No change
	Didanosine ^a	Reduce for eGFR < 60	No dose after dialysis
	Emtricitabine	Reduce for eGFR < 50	No change; dose after dialysis
	Lamivudine	Reduce for eGFR < 30	Reduce dose; dose after dialysis
	Stavudine ^a	Reduce for eGFR < 50	Reduce dose
	TDF	Reduce for eGFR < 50	Reduce dose; dose after dialysis
	Tenofovir alafenamide	Not recommended if eGFR < 15 and not on hemodialysis	No change
	Zidovudine	Reduce for eGFR < 15	Reduce dose; dose after dialysis
NNRTI	Doravirine	No change	Not studied
	Efavirenz	No change	No change
	Etravirine	No change	No change
	Nevirapine	No change eGFR ≥ 20, not recommended eGFR < 20	Additional 200 mg dose of immediate release formulation recommended after dialysis
	Rilpivirine	Use with caution in severe CKD	No data
Protease inhibitor	Atazanavir	No change	Depends on antiretroviral treatment status
	Darunavir	No change	No data
	Fosamprenavir	No data	No data
	Indinavir ^a	No change	No change
	Lopinavir	No data	Avoid once-daily dosing regimen
	Nelfinavir	No data	No data
	Saquinavir ^a	Not recommended for severe CKD	No data
	Tipranavir	Probably no change	No data
Fusion inhibitor	Enfuvirtide	No change	No change
Entry inhibitor (gp41)	Maraviroc ^Δ	eGFR < 30: dosing depends on concomitant medications	Dosing depends on concomitant medications
INSTI	Cabotegravir ^a	eGFR < 15: no data	No data
	Dolutegravir	No change; use caution in INSTI-experienced patients with severe renal impairment	No data
	Raltegravir	No change	Dose after dialysis
	Bictegravir	Not recommended	eGFR < 30 acceptable in dialysis if benefits outweigh risks
Pharmacokinetic enhancer	Cobicistat	No change	No data
	Ritonavir	For eGFR < 70, avoid combination with TDF	No data
CD4 postattachment inhibitor	Ibalizumab	No data	No data
gp120 attachment inhibitor	Fostemsavir	No change	No change

As of 2021, there are 30 antiretroviral medications and 24 antiretroviral medication combination formulations approved for use in the treatment of HIV disease in the United States.

^aNo longer available in the United States.

CCR5, C-C chemokine receptor 5; CKD, chronic kidney disease; CrCl, creatine clearance; eGFR, estimated glomerular filtration rate; gp, glycoprotein; HIV, human immunodeficiency virus; INSTI, integrase strand-transfer inhibitor; NNRTI, nonnucleoside reverse transcriptase inhibitor; NRTI, nucleoside reverse transcriptase inhibitor; TDF, tenofovir disoproxil fumarate.

(Table 58.2). These include protease inhibitors, fusion inhibitors (GP41 antagonist), an entry inhibitor (CCR5 antagonist), integrase inhibitor, integrase strand inhibitor (cabotegravir), and postattachment inhibitors, as well as a pharmacokinetic enhancer. Two long-acting injectable agents are available, cabotegravir and rilpivirine (NNRTI). Of these six classes, only protease inhibitors exhibit kidney toxicity. Indinavir was a formerly common cause of tubular injury but is rarely used at present and has been discontinued in the United States. Case series show that in individuals taking ritonavir-boosted atazanavir, urinary crystals composed of atazanavir affect 10% of subjects⁵⁶ and kidney stones occur at a rate of 2.4 cases per 100 patient-years of therapy.⁵⁷ Risk factors for nephrolithiasis include alkaline urine and kidney impairment; serum levels of atazanavir were not associated with stones.⁵⁸ Atazanavir has

been also associated with interstitial nephritis.⁵⁹ Atazanavir is no longer commonly used.

Tenofovir Disoproxil Fumarate

TDF is the most common cause of tubular injury in HIV+ individuals and manifests primarily as proximal tubular dysfunction with Fanconi syndrome. TDF is taken up by proximal tubular cells by one or more organic anion transporters and converted to the active drug that inhibits DNA repair and DNA replication in host cells and in HIV. TDF also causes mitochondrial enlargement and dysmorphic changes and is associated with cellular adenosine triphosphate (ATP) depletion.⁶⁰ A mild decrease in GFR has also been associated with TDF use, but usually it is not severe enough to require discontinuation of treatment.

TABLE 58.3 Kidney Toxicity of Antiretroviral Therapy

Antiretroviral Class	Antiretroviral Therapy	Kidney Effect	Clinical Recommendations
Protease inhibitors	Indinavir	Nephrolithiasis, crystalluria, dysuria, papillary necrosis, acute kidney injury, interstitial nephritis Ritonavir may increase toxicity of indinavir	Daily fluid intake of >2 L/day
	Atazanavir	Crystalluria, stones, interstitial nephritis	Daily fluid intake of >2 L/day
	Ritonavir	Increases levels of TDF	
Reverse transcriptase inhibitors	TDF	Kidney tubular damage: proximal tubular dysfunction, Fanconi syndrome, nephrogenic diabetes insipidus, acute tubular necrosis, acute kidney injury	Patients taking TDF should be monitored for signs of tubular dysfunction, including elevated serum creatinine, hypophosphatemia, low serum uric acid, acidosis, glycosuria, and proteinuria

TDF, Tenofovir disoproxil, fumarate.

Nonetheless, full recovery of kidney function may take months to years after stopping TDF.⁶¹ In some cases, recovery is partial or progresses to CKD.⁶² Risk factors for TDF-induced kidney toxicity include low CD4 cell count, prior kidney impairment, longer duration of therapy, and combined therapy with didanosine and protease inhibitors (lopinavir and ritonavir); the latter agents elevate TDF plasma levels, which, in turn, are associated with a higher risk of nephrotoxicity.⁶³

TDF is taken up into the kidney tubular epithelial cell from basolateral space by the organic anion transporter (OAT) 1 and OAT3 and is secreted into the tubular lumen by the multidrug resistance protein (MRP) 4. Mutations in *ABCC4*, encoding MRP4, are associated with increased plasma TDF concentrations and nephrotoxicity risk⁶⁴ and mutations in *ABCC10*, encoding MRP10, are associated with increased risk of phosphate wasting.⁶⁵ On the other hand, the use of MRP inhibitors such as NSAIDs, salicylates, and dipyridamole does not increase toxicity.⁶⁶

Tenofovir-alafenamide (TAF) has proven effective and well-tolerated. Like TDF, TAF is a prodrug but, unlike TDF, it is not a substrate for the OAT1 and OAT2 transporters and thus is not taken up by tubular cells. Currently available data suggest lower levels of kidney toxicity compared with TDF but longer trials will be required to confirm this.⁶⁷ TAF is also associated with less phosphaturia and better preserved bone mineralization in adults older than 60 years.

ANTIRETROVIRAL THERAPY DOSING IN CHRONIC KIDNEY DISEASE

Many antiretroviral medications are partially or completely eliminated by the kidney and require dosage adjustment in CKD. Certain drug classes, such as the protease inhibitors and the NNRTIs, are extensively metabolized by the liver and do not require dosage adjustment. Most of the NRTIs are excreted unchanged in the urine and require dosage adjustment, except for abacavir, which has substantial extrarenal biotransformation that requires little if any dosage adjustment. In uremia, drug dosages may be affected by altered gastric pH and altered volumes of distribution. Factors that influence dialyzability of antiretroviral medications relate to the properties of the particular dialysis membrane and molecular weight, degree of protein binding, molecular charge, and water solubility of the drug. Drugs that are substantially removed by hemodialysis should be taken after dialysis sessions. If the drug is removed in peritoneal dialysis effluent, the dose may have to be adjusted. Dosage recommendations (Table 58.3) in both hemodialysis and peritoneal dialysis are limited; there is a recommendation for NRTI dosing in patients undergoing venovenous hemofiltration.⁶⁸

END-STAGE KIDNEY DISEASE

With the increasing survival of HIV+ individuals with treatment and the declining cost of cART, the incidence of HIV-associated ESKD will likely increase worldwide, as it has in high-income countries. Currently, survival among dialysis-dependent HIV+ patients who are stable on cART is comparable to that among dialysis patients without HIV infection, and choice of dialysis modality (hemodialysis versus peritoneal dialysis) does not have an impact on survival.

Immunization schedules are the same as for non-HIV-negative dialysis patients.⁶⁹ In both CKD and HIV infection, anemia is independently associated with shorter survival. The response to recombinant erythropoietin (EPO) in HIV+ patients with ESKD is similar to that in HIV-negative patients.⁷⁰ Measurements of iron indices are complicated in HIV+ patients because levels of ferritin (an acute phase reactant) are often elevated in patients with HIV infection. Elevated soluble transferrin receptor concentrations are more reliable than serum ferritin to distinguish iron deficiency in inflammatory disease states.⁷¹

Hemodialysis

Strict adherence to universal precautions is the best form of prevention of HIV transmission in dialysis units, and there is no reason for these patients to be isolated. Even reprocessing of dialyzers from HIV+ patients does not place staff members at increased risk for infection, if necessary sterile precautions are undertaken. The risk of viral seroconversion after a needle-stick injury has been estimated as 6 in 1000 for HCV, 4 in 1000 for HBV, and 2 in 1000 for HIV.⁷² ART may reduce the risk of transmission after a needle stick injury, and postexposure prophylaxis for 28 days with a combination of two reverse transcriptase inhibitors and one protease inhibitor is recommended. Native arteriovenous fistulas are the preferred types of access for all patients with ESKD because of better patency rates and lower complication rates, compared with other access options. A recent study from South Africa showed similar patient survival outcomes for HIV+ and HIV-negative hemodialysis patients, except for increased incidence of tuberculosis in the HIV+ group.⁷³

Peritoneal Dialysis

HIV-1 remains infectious at room temperature in peritoneal dialysis (PD) exchange tubing for up to 48 hours and for up to 7 days in peritoneal effluent.⁷⁴ Dialysate should therefore be handled as a contaminated body fluid.⁷⁴ Sodium hypochlorite (50% solution) and household bleach (10% solution), each further diluted 1 to 512, are effective at inactivating HIV in dialysate. PD patients should be

instructed to pour dialysate into the home toilet and to dispose of dialysate bags and lines by tying them in plastic bags and disposing of the plastic bags in conventional home garbage.

Kidney Transplantation

The past few years have seen a surge of interest in kidney transplantation for HIV+ patients with ESKD. In a report of 150 kidney transplants from HIV-negative donors (both living and deceased) to HIV+ recipients,⁷⁵ acute rejection rates were higher than in other subjects, but HIV replication remained controlled. Similar outcomes were reported subsequently from other centers, including 25 HIV+ recipients who received 43 kidneys from HIV+ deceased donors in South Africa^{76,77} and 92 subjects in the United States.⁷⁸ With longer-term follow-up, compared with HIV-negative recipients, HIV+ recipients had a marginally increased risk of graft loss and no increased risk of death.⁷⁹ HIV+ pediatric recipients have also received kidneys from HIV-negative donors with good outcomes.⁸⁰

Interestingly, Canaud et al. reported that kidneys from HIV-negative donors became infected after transplantation into HIV+ recipients, with HIV RNA detected in podocytes and tubular cells.⁸¹ In 2013 the HOPE (HIV Organ Policy Equity) Act was passed by the U.S. Congress, which legalized research involving HIV+ living donors providing kidneys to HIV+ recipients, and studies with this population are being developed.⁸²

EVALUATION FOR KIDNEY DISEASE

HIV+ individuals should undergo regular evaluation for kidney disease, probably at least every 6 months and more frequently if kidney disease is suspected (Fig. 58.3). Early identification of kidney injury

disease offers the prospect of discovering kidney injury when it first appears, defining the nature of the process, removing provocative factors when possible (particularly medications), and initiating therapy to slow, halt, or possibly reverse the disease process.

Periodic testing should include a history of nephrotoxic medications and measurement of BP, serum creatinine, blood urea nitrogen, and urine protein/creatinine ratio. Three studies in HIV+ patients have compared the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equations against the gold standard iohalamate GFR: All three concluded that the CKD-EPI equation correlated well with measured GFR. When doses prescribed were compared with those recommended by the CKD-EPI eGFR, 6% to 19% of patients received an inappropriate dose; these subjects tended to be older and were more likely to have diabetes.

Most proteinuria in HIV+ patients is tubular in origin. In a study from London, 10% of subjects had proteinuria (defined as protein/creatinine ratio > 200 mg/g), most of which was not albumin; 17% had microalbuminuria, and 67% had increased excretion of the low molecular protein neutrophil gelatinase-associated lipocalin (NGAL).⁸³ Microalbuminuria, defined as 20 to 200 mg/g of creatinine, may indicate early glomerular disease (including in diabetic nephropathy, arterionephrosclerosis, HIV-associated glomerular disease), tubular disease, metabolic syndrome, or systemic inflammation. In six of seven South African patients with persistent microalbuminuria, HIVAN was present on kidney biopsy,⁸⁴ documenting that this disease may have an insidious presentation. Modestly elevated albuminuria may be persistent or intermittent and may reflect other processes, such as diabetes and hypertension. Indeed, in HIV+ individuals, microalbuminuria is associated with Framingham risk score,⁸⁵ endothelial dysfunction,⁸⁶ and intimal medial thickness and with all-cause mortality.⁸⁷

SELF-ASSESSMENT QUESTIONS

1. In individuals of African descent with HIV infection, genetic variants in the *APOL1* gene are associated with which of the following kidney biopsy diagnoses?
 - A. HIV-associated nephropathy (collapsing glomerulopathy) or focal segmental glomerulosclerosis
 - B. Membranous nephropathy
 - C. Immune complex glomerulonephritis
 - D. IgA nephropathy
2. An HIV-positive individual is receiving antiviral therapy and has suppressed viral replication. At a routine clinic visit, reduced serum phosphate and uric acid along with glycosuria are noted. Which of the following is the most likely explanation for these abnormalities?
 - A. Efavirenz
 - B. Allopurinol
 - C. Tenofovir disoproxil
 - D. Ritonavir
3. The clinician of the patient in question 2 was not provided with the laboratory results, and no changes were made in the clinical management of the patient. At the next visit 6 months later, the serum creatinine has risen and a kidney biopsy is performed. Which of the following findings would be most *specific* in clarifying the cause of the injury?
 - A. Arterionephrosclerosis
 - B. Focal global glomerulosclerosis
 - C. Tubular atrophy and interstitial nephritis
 - D. Giant mitochondria within the tubular cells

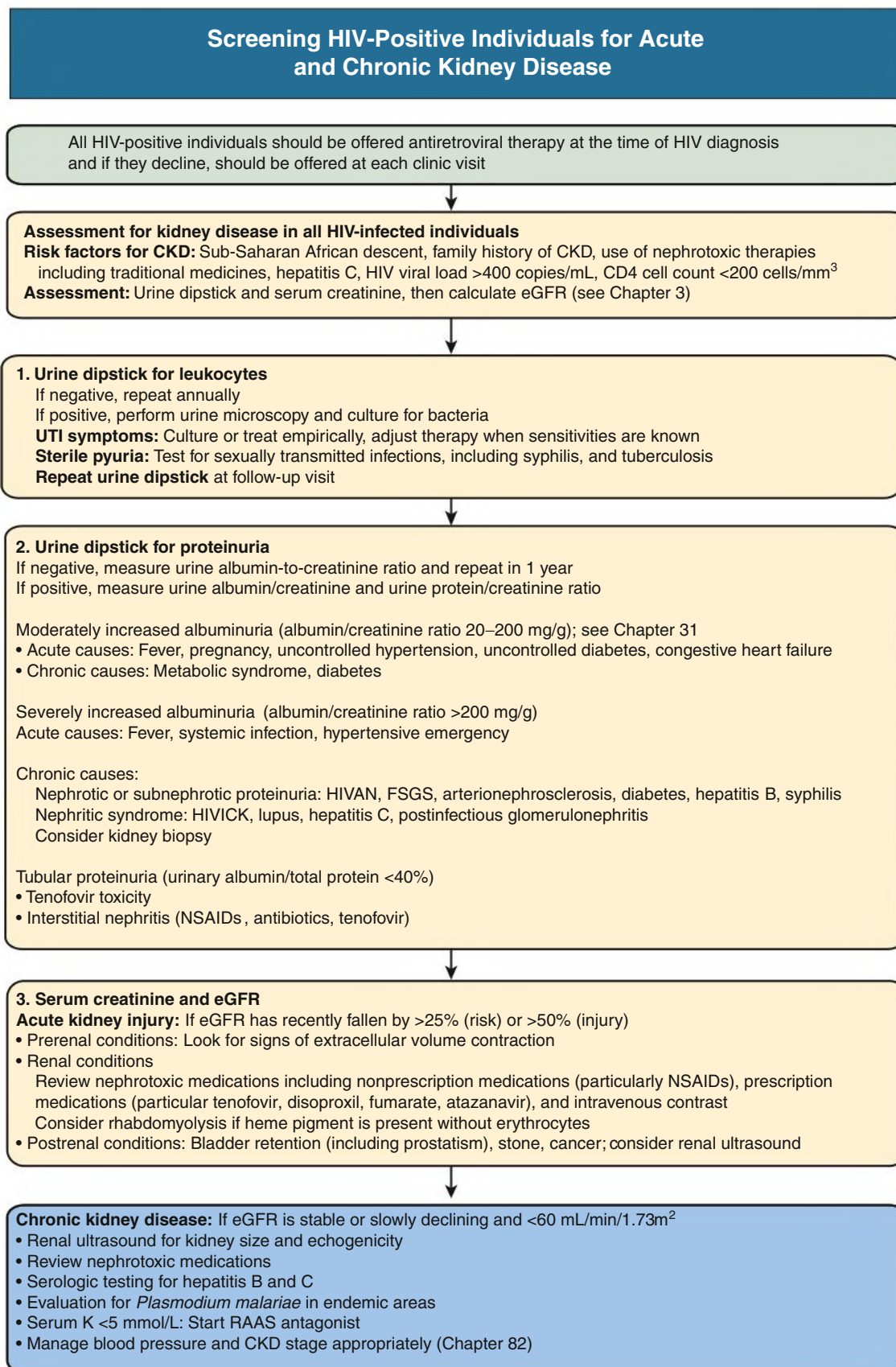


Fig. 58.3 Management Algorithm for Screening of Human Immunodeficiency Virus (HIV)-Positive Antiretroviral Therapy–Naïve Patients for Chronic Kidney Disease. Tuberculosis may be pulmonary or extrapulmonary. Antiproteinuric agents may be used in normotensive individuals with gradual up-titration of dose, depending on tolerance and severity of proteinuria. *CKD*, chronic kidney disease; *eGFR*, estimated glomerular filtration rate, calculated by the Cockcroft-Gault formula or the modified Modification of Diet in Renal Disease (MDRD) formula; *FSGS*, focal segmental glomerulosclerosis; *HIVAN*, HIV-associated nephropathy; *HIVICK*, HIV-associated immune complex kidney disease; *NSAID*, nonsteroidal anti-inflammatory drug; *RAAS*, renin-angiotensin-aldosterone system; *STI*, sexually transmitted infection; *UTI*, urinary tract infection.

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